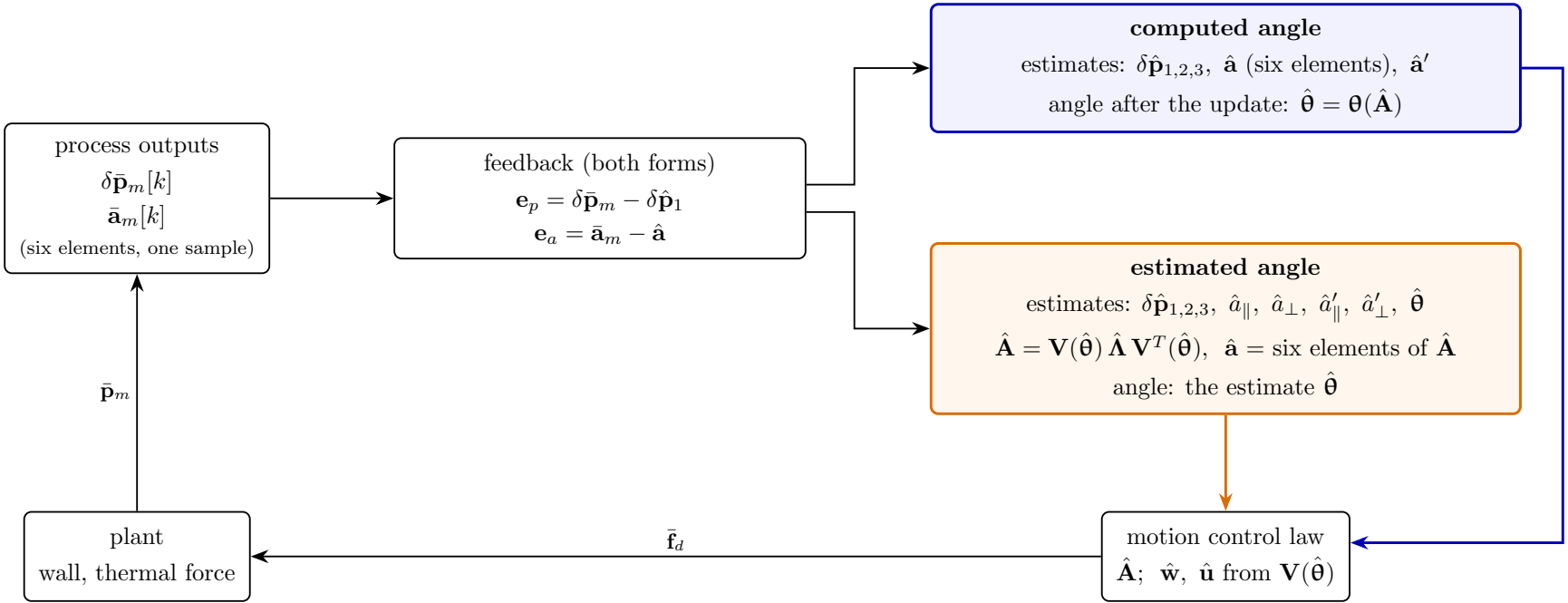

process outputs
$\delta\bar{\mathbf{p}}_m[k]$
$\bar{\mathbf{a}}_m[k]$
(six elements, one sample)
feedback (both forms)
$\mathbf{e}_p = \delta\bar{\mathbf{p}}_m - \delta\hat{\mathbf{p}}_1$
$\mathbf{e}_a = \bar{\mathbf{a}}_m - \hat{\mathbf{a}}$
computed angle
estimates: $\delta\hat{\mathbf{p}}_{1,2,3}$, $\hat{\mathbf{a}}$ (six elements), $\hat{\mathbf{a}}'$
angle after the update: $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}(\hat{\mathbf{A}})$
estimated angle
estimates: $\delta\hat{\mathbf{p}}_{1,2,3}$, $\hat{a}_\parallel$, $\hat{a}_\perp$, $\hat{a}'_\parallel$, $\hat{a}'_\perp$, $\hat{\boldsymbol{\theta}}$
$\hat{\mathbf{A}} = \mathbf{V}(\hat{\boldsymbol{\theta}})\,\hat{\boldsymbol{\Lambda}}\,\mathbf{V}^T(\hat{\boldsymbol{\theta}})$, $\hat{\mathbf{a}}$ = six elements of $\hat{\mathbf{A}}$
angle: the estimate $\hat{\boldsymbol{\theta}}$
$\bar{\mathbf{p}}_m$
plant
wall, thermal force
$\bar{\mathbf{f}}_d$
motion control law
$\hat{\mathbf{A}}$; $\hat{\mathbf{w}}$, $\hat{\mathbf{u}}$ from $\mathbf{V}(\hat{\boldsymbol{\theta}})$